

Selected Topics Of Living Systems HW1

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Technion

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1 Interfacial Tension

Consider the Ginzburg-Landau free energy in 1D

$$F[p] = A \int dx \left(\frac{a}{2} p^2 + \frac{b}{4} p^4 + \frac{K}{2} p'^2 \right) \quad (1)$$

where A is the cross section (area) and where $a < 0$ and $b, K > 0$. The equilibrium value of p in the homogenous case is $p = \pm p_{eq}$ with $p_{eq} = \sqrt{-a/b}$. Given the boundary conditions $p(x \rightarrow \pm\infty) = \pm p_{eq}$, we have found the equilibrium profile $p_0(x) = p_{eq} \tanh(x/\xi)$, where $\xi = \sqrt{2K/|a|}$. Calculate the interfacial tension $\gamma = (F[p_0] - F[\pm p_{eq}]) / A$. Show that

$$\gamma = \int dx K p_0'^2 = \sqrt{\frac{-8Ka^3}{9b^2}} \quad (2)$$

First we find

$$p_0' = \frac{dp_0}{dx} = p_{eq} \frac{d \tanh(x/\xi)}{dx} = \frac{p_{eq}}{\xi \cosh^2(x/\xi)} = \frac{p_{eq}}{\xi} \text{sech}^2(x/\xi) \quad (3)$$

therefore

$$p_0'^2 = \frac{p_{eq}^2}{\xi^2} \text{sech}^4(x/\xi) = -\frac{a}{b\xi^2} \text{sech}^4(x/\xi) \quad (4)$$

Next we directly calculate the interfacial tension

$$\gamma = (F[p_0] - F[\pm p_{eq}]) / A \quad (5)$$

$$= \left(A \int dx \left(\frac{a}{2} p_0^2 + \frac{b}{4} p_0^4 + \frac{K}{2} p_0'^2 \right) - A \int dx \left(\frac{a}{2} p_{eq}^2 + \frac{b}{4} p_{eq}^4 + \frac{K}{2} p_{eq}'^2 \right) \right) / A \quad (6)$$

$$= \int dx \frac{a}{2} p_0^2 + \frac{b}{4} p_0^4 + \frac{K}{2} p_0'^2 - \frac{a}{2} p_{eq}^2 - \frac{b}{4} p_{eq}^4 \quad (7)$$

$$= \frac{1}{2} \int dx a p_0^2 + \frac{b}{2} p_0^4 + K p_0'^2 - a p_{eq}^2 - \frac{b}{2} p_{eq}^4 \quad (8)$$

$$= \frac{1}{2} \int dx a (p_0^2 - p_{eq}^2) + \frac{b}{2} (p_0^4 - p_{eq}^4) + K p_0'^2 \quad (9)$$

$$= \frac{1}{2} \int dx a p_0^2 + \frac{a^2}{2b} + \frac{b}{2} p_0^4 - \frac{a^2}{2b} + K p_0'^2 \quad (10)$$

$$= \frac{1}{2} \int dx a p_0^2 + \frac{b}{2} p_0^4 + K p_0'^2 \quad (11)$$

$$= \frac{1}{2} \int dx a p_{eq}^2 \tanh^2(x/\xi) + \frac{b}{2} p_{eq}^4 \tanh^4(x/\xi) + K \frac{p_{eq}^2}{\xi^2} \text{sech}^4(x/\xi) \quad (12)$$

$$= \frac{1}{2} \int dx a p_{eq}^2 (1 - \text{sech}^2(x/\xi)) + \frac{b}{2} p_{eq}^4 (1 - \text{sech}^2(x/\xi))^2 + K \frac{p_{eq}^2}{\xi^2} \text{sech}^4(x/\xi) \quad (13)$$

$$\gamma = \frac{1}{2} \int dx \, a p_{eq}^2 (1 - \text{sech}^2(x/\xi)) + \frac{b}{2} p_{eq}^4 (1 - \text{sech}^2(x/\xi))^2 + K \frac{p_{eq}^2}{\xi^2} \text{sech}^4(x/\xi) \quad (14)$$

$$= \int dx \, a p_{eq}^2 + b p_{eq}^4 + p_{eq}^2 \text{sech}^2(x/\xi) \left(\frac{K}{\xi^2} \text{sech}^2(x/\xi) - a - b p_{eq}^2 \right) \quad (15)$$

$$= \int dx \, \frac{-a^2}{b} + \frac{a^2}{b} - \frac{a}{b} \text{sech}^2(x/\xi) \left(\frac{K}{\xi^2} \text{sech}^2(x/\xi) \right) \quad (16)$$

$$= \int dx \, -\frac{aK}{b\xi^2} \text{sech}^4(x/\xi) \quad (17)$$

$$= \int dx \, K p_0'^2 \quad (18)$$

next to find the actual value of γ we calculate the integral itself

$$\gamma = \int dx \, K p_0'^2 = \int dx \, -\frac{aK}{b\xi^2} \text{sech}^4(x/\xi) \quad (19)$$

$$= -\frac{aK}{b\xi^2} \int dx \, \text{sech}^4(x/\xi) \quad (20)$$

$$= [u = \tanh(x/\xi)] = -\frac{aK}{b\xi^2} \xi \int_{-1}^1 du \, (1 - u^2) \quad (21)$$

$$= -\frac{aK}{b\xi^2} \xi \cdot \frac{4}{3} \quad (22)$$

$$= -\frac{4aK}{3b\xi} \quad (23)$$

$$= -\frac{4aK}{3b\sqrt{\frac{2K}{|a|}}} \quad (24)$$

$$= \sqrt{\frac{-8Ka^3}{9b^2}} \quad (25)$$

as required.

2 FDT

The fluctuation-dissipation theorem (FDT) related the correlation function $C(t)$ with the response function $\chi(t)$ according to

$$\frac{\partial C}{\partial t} = -k_B T \chi(t) \quad t > 0 \quad (26)$$

we define the fourier transformations

$$\tilde{C}(\omega) = \int_{-\infty}^{\infty} dt C(t) e^{i\omega t} \quad (27)$$

$$\tilde{\chi}(\omega) = \int_0^{\infty} dt \chi(t) e^{i\omega t} \equiv \tilde{\chi}'(\omega) + i\tilde{\chi}''(\omega) \quad (28)$$

where $\tilde{\chi}'(\omega)$ and $\tilde{\chi}''(\omega)$ are the real and imaginary parts of the transformed response function, respectively. Show that $C(t)$ is an even function and derive FDT in the frequency domain,

$$\tilde{C}(\omega) = \frac{2k_B T}{\omega} \tilde{\chi}''(\omega) \quad (29)$$

In the lecture we saw that C does not depend on the explicit times but rather only on the time that has passed, therefore we can rewrite

$$C(t) = \langle \delta x(0) \delta x(t) \rangle \rightarrow C(t) = \langle \delta x(-t/2) \delta x(t/2) \rangle \quad (30)$$

which we can immediately notice that, assuming the system is invariant to time reversal,

$$C(-t) = \langle \delta x(t/2) \delta x(-t/2) \rangle = C(t) \quad (31)$$

therefore C is even. Next, we look at

$$\frac{\partial C}{\partial t} = -k_B T \chi(t) \quad t > 0 \quad (32)$$

and notice that since C is even w.r.t time, its first time derivative must be odd. This means χ must be odd as well and only depend on the absolute amount of time between t and 0 :

$$\frac{\partial C}{\partial t} = -k_B T \chi(|t|) \text{sgn}(t) \quad (33)$$

we'll take the fourier transform of the derivative

$$\int_{-\infty}^{\infty} dt \frac{\partial C}{\partial t} e^{i\omega t} = -k_B T \int_{-\infty}^{\infty} dt \chi(|t|) \text{sgn}(t) e^{i\omega t} \quad (34)$$

$$= -k_B T \left(\int_0^{\infty} dt \chi(|t|) e^{i\omega t} - \int_{-\infty}^0 dt \chi(|t|) e^{i\omega t} \right) \quad (35)$$

$$= -k_B T (\tilde{\chi}(\omega) - \tilde{\chi}(-\omega)) \quad (36)$$

since χ is real, we can use the identity $\tilde{\chi}(-\omega) = \tilde{\chi}^*(\omega)$ and find

$$\tilde{\chi}(\omega) - \tilde{\chi}(-\omega) = \tilde{\chi}(\omega) - \tilde{\chi}^*(\omega) \quad (37)$$

$$= 2i\Im(\tilde{\chi}(\omega)) \quad (38)$$

$$= 2i\tilde{\chi}''(\omega) \quad (39)$$

as for the other side of the equation we can use another fourier transform identity to find

$$\int_{-\infty}^{\infty} dt \frac{\partial C}{\partial t} e^{i\omega t} = i\omega \tilde{C}(\omega) \quad (40)$$

therefore

$$i\omega \tilde{C}(\omega) = 2ik_B T \tilde{\chi}''(\omega) \quad (41)$$

cancelling out terms we find

$$\tilde{C}(\omega) = \frac{2k_B T}{\omega} \tilde{\chi}''(\omega) \quad (42)$$

as required.